

Green Campus Sustainability Intelligence Platform

Simple user guide for terms, formulas, and key dashboard values

Parameter	Value
Emission factor	0.82 kg CO2 per kWh
Energy cost	8.0 per kWh
Energy benchmark	100000
Water benchmark	20000
Waste benchmark	30000
Green Index weights	Energy 0.5, Water 0.3, Waste 0.2
Alert thresholds	15% standard, 30% high

1. What this system is

- Green Campus is a dashboard that helps a campus track electricity, water, waste, solar, carbon, and future planning in one place.
- It is meant for a user who wants to understand the current situation clearly and make better decisions for the next period.

2. What the main values mean

- Gross Energy means all electricity used before solar support is removed.
- Solar means electricity produced by the solar system.
- Net Energy means electricity still needed from the grid after using solar.
- Water means total water use in the selected scope.
- Waste means total waste in the selected scope.
- Carbon means estimated emissions linked to the remaining grid electricity.
- Green Index means the final overall sustainability score. Higher is better.

3. Basic formulas used by the dashboard

- Net Energy = $\max(\text{Gross Energy} - \text{Solar}, 0)$.
- Export to Grid = $\max(\text{Solar} - \text{Gross Energy}, 0)$.
- Carbon = Net Energy x carbon factor.
- Gross Carbon = Gross Energy x carbon factor.
- Solar Avoided Carbon = $\min(\text{Solar}, \text{Gross Energy}) \times \text{carbon factor}$.

4. Green Index in simple words

- The system first checks whether energy, water, and waste are high or low compared with their benchmark values.
- Then it combines those three results using weights.
- Energy has the biggest effect, water the next, and waste the least.
- After that, the system converts the result into a score out of 100.
- If energy, water, and waste are lower and more efficient, the Green Index becomes higher.

5. Green Index step by step

- Step 1: Compare Net Energy with the energy benchmark.
- Step 2: Compare Water with the water benchmark.
- Step 3: Compare Waste with the waste benchmark.
- Step 4: Multiply those three results by their weights.
- Step 5: Add them to get one combined efficiency pressure value.
- Step 6: Convert that pressure into the final Green Index score.
- Simple formula: $\text{Green Index} = (1 - \text{combined efficiency pressure}) \times 100$.

6. Important terms in simple language

- Benchmark means a reference value used to judge whether usage is low or high.
- Weighted Efficiency means the combined effect of energy, water, and waste after giving each one different importance.
- Forecast Range means the system shows a likely lower and upper band instead of pretending one exact future number is guaranteed.
- Occupancy Effect means future demand changes because campus activity changes from month to month.
- Export Potential means extra solar energy that may remain after campus demand is met.

7. What occupancy values mean

- 1.0 means normal or full campus activity.
- 0.9 means slightly reduced activity.
- 0.5 means partial activity or vacation-like activity.
- 0.2 means very low campus activity.
- These are activity multipliers, not exact percentages.

8. Forecast and seasonal logic in simple words

- The system looks at past data and groups it by daily, monthly, yearly, or seasonal view.
- Then it uses those patterns to estimate future values.
- Lower occupancy months like June and July usually reduce expected energy and water demand.
- At the same time, solar may cover a larger share of demand during low-activity months.

9. Alerts, recommendations, and action board

- Alerts help the user notice spikes, unusual changes, and possible problem areas.
- Recommendations suggest what action could improve energy, cost, carbon, or resource behavior.
- The Action Board lets the user mark each recommendation as suggested, planned, in progress, or completed.

10. Import, export, and reset

- Import Excel loads a fresh workbook into the system.
- Export Excel downloads the current system data as a workbook.
- Reset Data clears the current records so a new dataset can be loaded.